AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

T318
Applied Network Security

Final Examination
Fall Semester 2025/2026

Date: 20/Jan./2026

Number of Exam Pages: 6 (including this cover sheet)	**Time Allowed:** 3 Hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: MULTIPLE CHOICE QUESTIONS [10 Marks]

Question 1: (10 marks)

1. Which of the following best distinguishes transposition from substitution?
A. Transposition changes symbols; substitution changes order
B. Transposition changes order; substitution changes symbols
C. Both change symbols only
D. Both use public keys

2. Which of the following is a transposition cipher?
A. Caesar cipher
B. Vigenère cipher
C. Rail Fence cipher
D. Monoalphabetic cipher

3. Which classical cipher is also known as a polyalphabetic cipher?
A. Caesar cipher
B. Hill cipher
C. Vigenère cipher
D. Rail Fence cipher

4. In a secure communication system (e.g., HTTPS), which statement is TRUE?
A. Only asymmetric encryption is used
B. Only symmetric encryption is used
C. Asymmetric encryption is used to exchange symmetric keys
D. Symmetric encryption is used to generate public keys

5. Why do block ciphers often require modes of operation (e.g., CBC, CTR)?
A. To generate public keys
B. To handle data larger than one block securely
C. To reduce key size
D. To avoid decryption

6. Which cryptographic structure is used by DES?
A. Substitution–Permutation Network (SPN)
B. Feistel network
C. Stream cipher
D. Hash-based structure

7. Which statement best describes the relationship between X.509 certificates and public keys?
A. Certificates store the public key along with identifying information and CA signature
B. Certificates store only private keys
C. Certificates are used to encrypt the CA's private key
D. Certificates are unrelated to public keys

8. What is the order of AES operations in a standard round?
A. ShiftRows → SubBytes → MixColumns → AddRoundKey
B. SubBytes → ShiftRows → MixColumns → AddRoundKey
C. MixColumns → ShiftRows → SubBytes → AddRoundKey
D. SubBytes → MixColumns → ShiftRows → AddRoundKey

9. In IPsec Tunnel mode, which of the following is TRUE?

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A. Only the payload of the IP packet is encrypted
B. The entire IP packet (header + payload) is encrypted and a new IP header is added
C. No encryption is performed
D. Only TCP/UDP headers are encrypted

10. What is the main difference between misuse-based IDS (signature-based) and anomaly-based IDS?
A. misuse-based detects known attacks; anomaly-based detects deviations from normal behavior
B. misuse-based is faster than anomaly-based
C. Anomaly-based requires no learning phase
D. misuse-based cannot detect any attacks

PART 2: SHORT ANSWERS QUESTIONS [16 Marks]

Question 2: (8 marks)

A company has four departments: HR, Finance, IT, and Sales. Each department has several employees who need to communicate securely over the company network.

The network administrator has decided to implement public key cryptography and uses a Certificate Authority (CA) to issue digital certificates to all employees. Each employee has a public/private key pair.

1. Explain how employees can establish a secure session key using certificates issued by the CA.

2. If there are 10 employees on the network, how many certificates need to be issued by the CA?

Suppose Alice wants to send a message to Bob.

3. Describe the steps Alice should follow to verify Bob's public key using his certificate.

4. What security properties does verifying a certificate provide?

5. Which of the following statements best describes the main difference between a Key Distribution Center (KDC) and a Certificate Authority (CA)?
A. A KDC issues digital certificates, while a CA distributes symmetric session keys.
B. A KDC distributes symmetric session keys and must be fully trusted, while a CA verifies public keys through certificates without directly handling session keys.
C. A KDC is used only in large networks, while a CA is used only in small networks.
D. A KDC and a CA perform the same function; the terms are interchangeable.

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Question 3: (8 marks)

Arab Open University has developed an online **Student Information System (SIS)** that provides services to different users.

- **Students**
- **Tutors**
- **Course coordinators**
- **System Administrators**

The system stores sensitive data such as:

- Student grades
- Course materials
- Exam questions
- User credentials

To protect the system, proper **authentication** and **access control mechanisms** must be designed and implemented.

1. Define authentication and explain why it is important in information systems.
2. Identify two authentication methods suitable for the SIS. For each method state one advantage and one limitation
3. Recommend the best authentication method for:
   - Students
   - System Administrators

   Justify your answer.
4. Explain what could go wrong if authentication is weak.
5. Identify and briefly explain **two common types of password attacks**.
6. Define **password salting**.
7. Define **password peppering**.
8. A database containing hashed passwords is leaked. Explain how **salting** limits the damage of the breach.

PART 3: PROBLEM SOLVING [24 Marks]

Question 4: (6 Marks)

Alice wants to send a confidential message to Bob using RSA encryption.
Bob chooses the following parameters:
Prime number p=3
Prime number q=11

Q1. What is the value of the RSA modulus n?
A. 33
B. 40
C. 30
D. 35

Q2. What is the value of Euler's totient function φ(n)?
A. 40
B. 50
C. 30
D. 20

Q3. What is the value of the private key d if the public key was e=3?
A. 7
B. 13
C. 5
D. 17

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Q4. Which formula is used for encryption?

A. $C = M \times e \bmod n$

B. $C = M^e \bmod n$

C. $C = M + e \bmod n$

D. $C = M^n \bmod e$

Q5. What is the Ciphertext of the message M=4?

A. 4

B. 16

C. 31

D. 32

Alice and Bob want to agree on a shared secret using Diffie-Hellman.
They agree on public parameters:
p=**11**(prime),
g=**2** (primitive root modulo 11)
Alice chooses private key a=**2**
Bob chooses private key b=**4**

6. Compute the shared secret key K=?

Question 5: (9 marks)

Alice wants to send a message $M$to Bob over an insecure channel. They want to ensure:

1. **Message integrity** – the message is not modified.
2. **Authentication** – Bob can verify that the message is from Alice.

Alice and Bob share a **secret key** K. They decide to use:

- A **hash function** $H()$
- A **Message Authentication Code (MAC)**

Message: $M = "HELLO"$
Secret key: $K = 3$
Hash function $H(M)$ = **sum of ASCII values of characters modulo 10**

Character	0	1	2	3	4	5	6	7	8	9
ASCII Code	48	49	50	51	52	53	54	55	56	57

Character	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
ASCII Code	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90

1. Compute the hash of the message $H(M)$.
2. Compute the MAC using the secret key K.
   Bob receives the message $M$and the MAC.
3. Show how Bob verifies the message integrity and authenticity using the shared key K.
4. What happens if an attacker changes one character in M? How does the MAC help detect it?
5. Explain how using a **MAC** provides **authentication** of the sender.

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6. What is the difference between using a **hash function alone** and using a **MAC** for message authentication?
7. Compute the hash of the message M'=" AOU16", compare it with the hash of M, what we call this situation?

Question 6: (5 Marks)

A packet filtering firewall has the following rule set.

No.	Source Address	Dest. Address	Source port	Dest. port	Protocol	Direction	Action
1	External	Internal	>1023	80	TCP	In	Permit
2	Internal	External	80	>1023	TCP	Out	Permit
3	Internal	External	>1023	80	TCP	Out	Permit
4	External	Internal	80	>1023	TCP	In	Permit
5	Any	Any	Any	Any	Any	Either	Deny

>1023 means any port except the well-known ports (reserved ports)

Someone from the outside world attempts to open a connection from port 80 on a remote host (20.1.2.3) to a server on port 1025 on one of your local hosts (136.16.2.4), to carry out an attack. The packets exchanged are as follows:

No.	Source Address	Dest. Address	Source port	Dest. port	Protocol	Direction	Action
1	20.1.2.3	136.16.2.4	80	1025	TCP	In	
2	136.16.2.4	20.1.2.3	1025	80	TCP	Out	

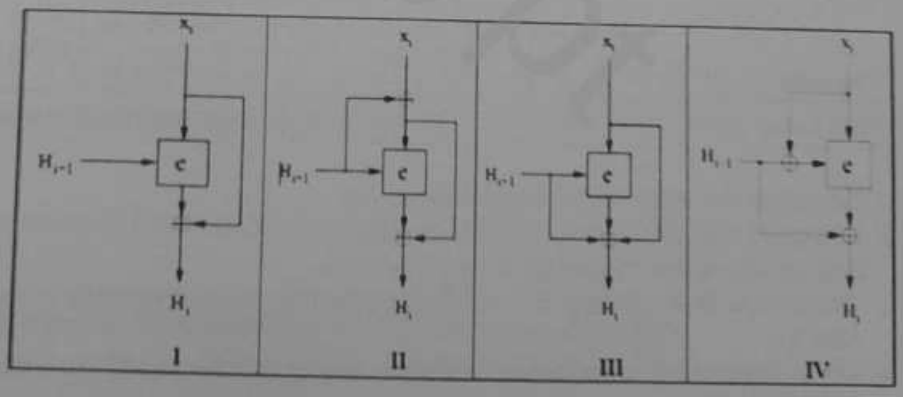
1. Will the attack succeed?

2. Explain which rule(s) is (are) used to permit or deny each packet by the firewall?

3. In case the attack succeeds, what solution do you propose to mitigate such attack?

Question 7: (4 Marks)

The following block diagrams represent four hash functions built from a block cipher e(), write the expression of each hash function.



End of questions

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